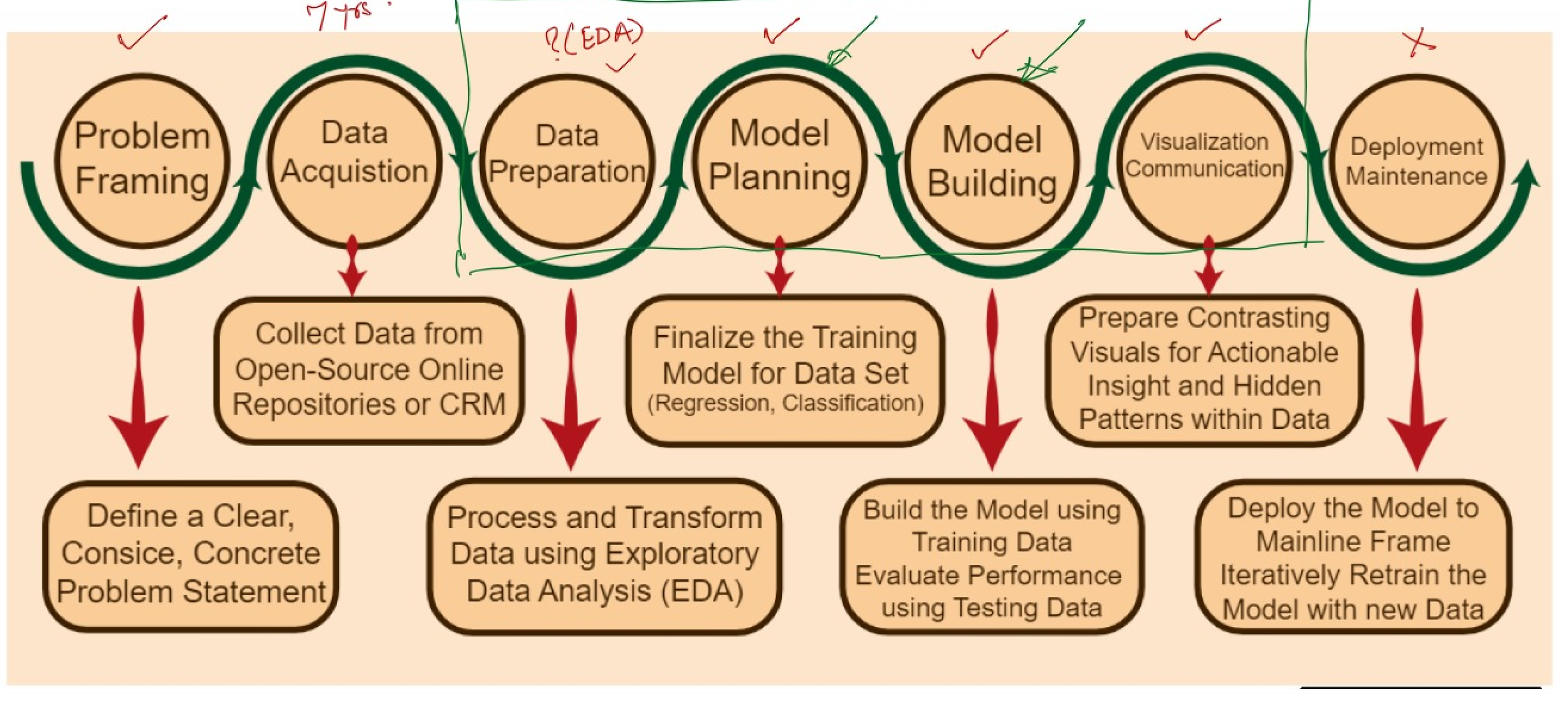
DS LIFECYCLE.



Todays class →

Python — Series of Examples (Regression) — DS Lifecycle.

ML algo (Regression, Classification)

→ Use the most appropriate tool / method. algo.

- Excel.
- Python — Sklearn.
- Statsmodels.

independent var. (feature) — dependent var. (response var.)

x	y
0	13.92
0.1	0.33095
0.2	-1.4562
0.3	18.23855
0.4	-12.3848
0.5	-12.1263
0.6	10.1942
0.7	5.19655
0.8	-11.7392
0.9	-1.83305
1	8.615

Goal: solve a problem

$y = f(x)$

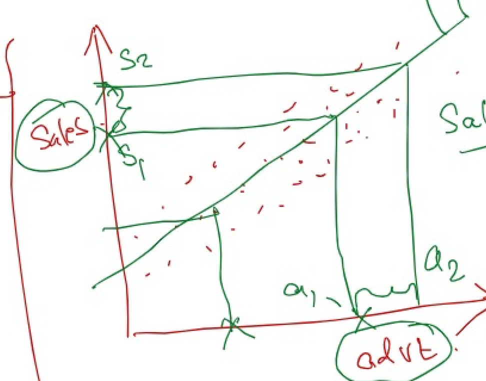
- have data
- Do not have relationship.
- relation between y & x.
- Given y & x can we find out the reln between them?

↓ pattern.

$s = ut + \frac{1}{2}at^2$

$y = f(x)$.

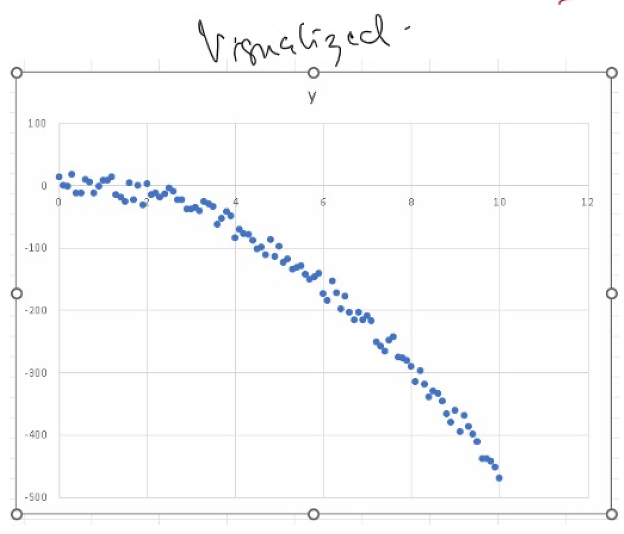
y: Sales	x: advt. Expense.
—	—



Sales = f(advt).

$\Delta(a) \rightarrow ?\ \Delta S$

Visualized.



$y = f(x)$.

Is linear Regression applicable?

Summary Statistics -

EDA

Mean	-150.0218564
Standard Error	14.16250244
Median	-113.26905
Mode	#N/A
Standard Deviation	142.331388
Sample Variance	20258.22401
Kurtosis	-0.823625901
Skewness	-0.655699944
Range	487.37855
Minimum	-469.14
Maximum	18.23855
Sum	-15152.2075
Count	101

Regression

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.962312357
R Square	0.926045072
Adjusted R Square	0.925298053
Standard Error	38.90152667
Observations	101

ANOVA — stats.

	df	SS	MS	F	Significance F
Regression	1	1876003	1876003	1239.653	8.58E-58
Residual	99	149819.5	1513.329		
Total	100	2025822			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	83.70929674	7.684558	10.89318	1.24E-18	68.46147	98.95713	68.46147	98.95713
x	-46.74623063	1.327689	-35.2087	8.58E-58	-49.3807	-44.1118	-49.3807	-44.1118

$y = f(x)$.

$y = \beta_0 + \beta_1 x$.

intercept. coefficient.

$\boxed{y = 83.71 - 46.75 . x}$

Eqn. of the regression line.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.996858
R Square	0.993726
Adjusted R	0.993598
Standard E	11.388
Observatic	101

Higher the better.

ANOVA

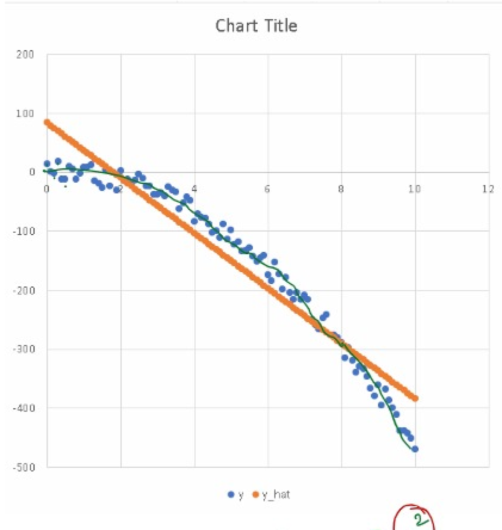
	df	SS	MS	F	Significance F
Regression	2	2013113	1006557	7761.452	1.2E-108
Residual	98	12709.29	129.6866		
Total	100	2025822			

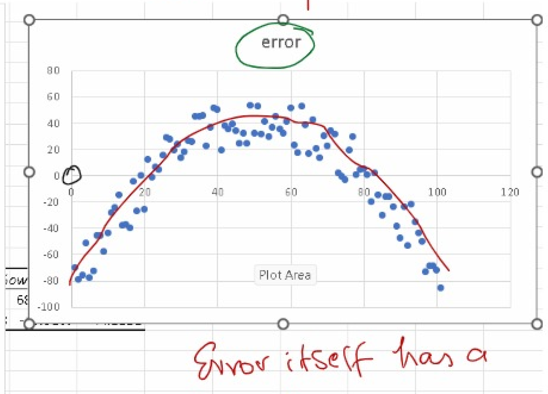
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	3.733571	3.333224	1.120108	0.265405	-2.8811	10.34825	-2.8811	10.34825
x1	1.723906	1.540524	1.119039	0.265859	-1.33321	4.781026	-1.33321	4.781026
x2	-4.84701	0.149069	-32.5153	2.65E-54	-5.14284	-4.55119	-5.14284	-4.55119

Should be less than 0.05.

$\boxed{y = 3.73 + 1.72 x_1 - 4.84 x_2}$

"actual - predicted"





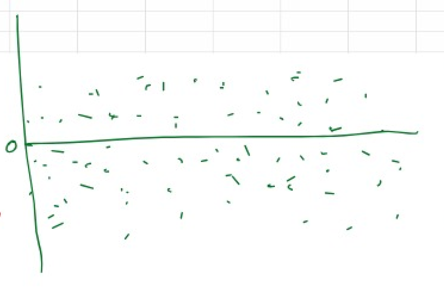
Error itself has a trend

NOT GOOD ✗

$y = \beta_0 + \beta_1 x + \beta_2 x^2$

Polynomial Regression.

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, $x_2 = x^2$

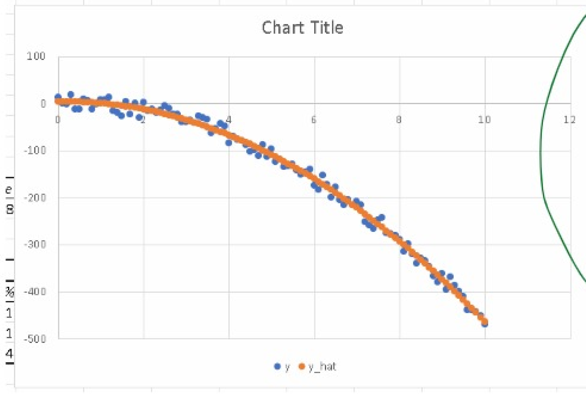


Random.

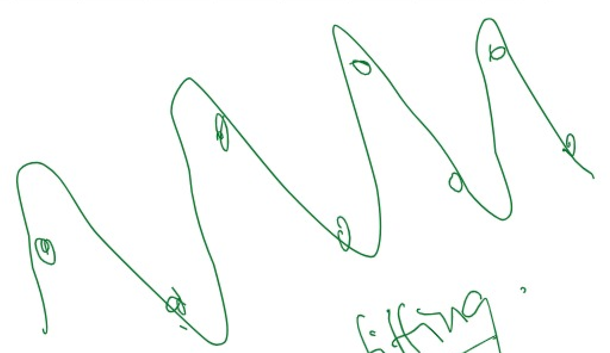
Mean is zero.

Normally distributed.

Taylor.



VERY GOOD!



Overfitting.

- Durbin Watson test
- Jarque - Bera

Bias Variance tradeoff.

$y = 3.73 + 1.72 x_1 - 4.84 x_2$

$s = ut + \frac{-1 \times 9}{2} t^2$

$y = f(x)$